

GLMs for Applied Research

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Session 2: Logistic Regression — Theory & Fitting in R

Topics we will cover in this session

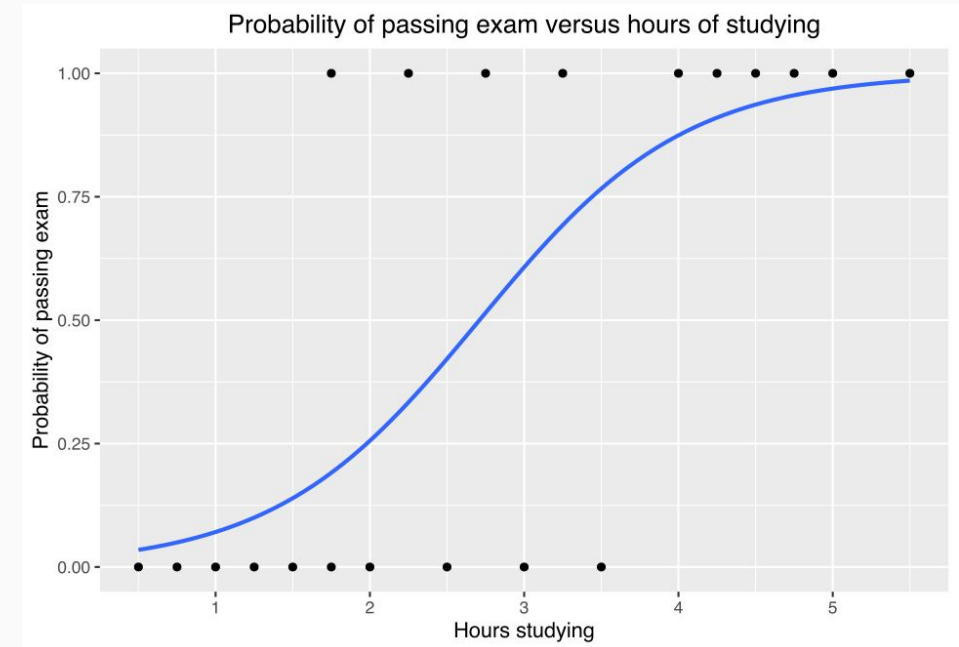
- Logistic regression model
- Odds ratios – interpreting coefficients
- Model fit in R and assessment
- Model evaluation
- Common pitfalls

Recap: Why not using linear regression

- Fitting a straight line to a 0/1 outcome produces predicted values outside $[0, 1]$ — not valid probabilities
- Residuals are inherently non-normal and heteroscedastic for binary data
- We need a model that:
 - Constraints predictions to $(0, 1)$
 - Handles the variance structure of binary outcomes correctly

The logistic (sigmoid) function

- Model the **probability** of the outcome, but pass a linear combination of predictors through a function that squashes it into (0, 1)
- $p = 1 / (1 + e^{(-z)})$, where z is the linear predictor ($\beta_0 + \beta_1 x_1 + \dots$)
- S-shaped curve: near-flat at the extremes, steep in the middle
- No matter what values the predictors take, output is always a valid probability



Log-odds (logit) link

- Odds = $p / (1 - p)$ — how much more likely success is than failure
- Logit = $\log(\text{odds}) = \beta_0 + \beta_1 x_1 + \dots \rightarrow$ **this is linear in the predictors**
- Logistic regression is a linear model, just on the log-odds scale instead of the probability scale
- Coefficients (β) are changes in log-odds per unit increase in x — not directly probabilities

Logistic regression: The model

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

The model **does not predict probability directly** but rather predicts the **log odds**, which can then be transformed back:

$$p = \frac{e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}}{1 + e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}}$$

- No closed-form solution like OLS — coefficients are estimated by **maximum likelihood**
- Idea: find the β values that make the observed data most probable under the model
- Solved iteratively (iteratively reweighted least squares) — R does this for you

Interpreting coefficients: Odds Ratio (OR)

Exponentiating a coefficient gives the **odds ratio** for a one-unit increase in that predictor, *holding all other predictors constant*.

$$OR_j = e^{\beta_j}$$

β value	OR = e^{β}	Interpretation
0	1.0	No effect on odds
0.69	2.0	Doubles the odds
-0.51	0.6	Reduces odds by 40%

- For a continuous predictor: OR shows the change in odds for a 1-unit increase in X.
- For a categorical predictor: OR shows the difference in odds between the category and the reference level.

Detailed interpretation during the hands-on session.

Inference on coefficients

- Wald test: $z = \hat{\beta} / SE(\hat{\beta})$ — analogous to t-test in OLS
- Likelihood Ratio Test (LRT): compares nested models via -2 log-likelihood difference; more reliable than Wald for small samples.

Model fit

- Null deviance: fit of intercept-only model
- Residual deviance: fit of the current model
- Reduction in deviance $\approx$ explained variation (analogous to R^2)
- $AIC = -2\ell + 2p$ — for comparing non-nested models (lower = better)

Goodness-of-fit

- Hosmer-Lemeshow test: groups observations by predicted probability deciles and tests whether observed vs. predicted counts match
- Calibration plots: predicted probability vs. observed event rate

Classification metrics

At a chosen threshold, e.g., $p > 0.5$:

- Sensitivity = $TP / (TP + FN)$
Specificity = $TN / (TN + FP)$
PPV = $TP / (TP + FP)$
NPV = $TN / (TN + FN)$
Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

— all derived from the confusion matrix. →

		Actual Values	
		Positive	Negative
Predicted Values	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

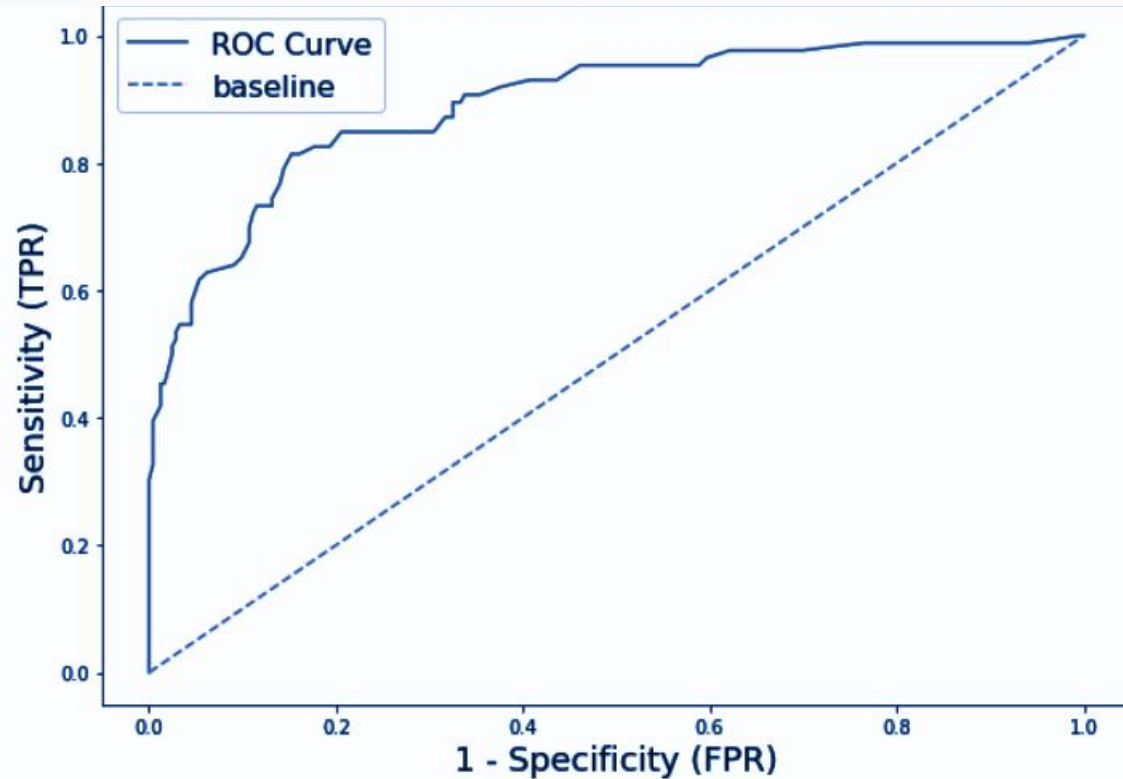
Model Performance

A model can be statistically significant but practically useless.

Discrimination: Can the model separate cases from non-cases?

- **ROC curve:** plots Sensitivity vs. $(1 - \text{Specificity})$ across all classification thresholds
- **AUC (C-statistic):** probability that a randomly chosen case scores higher than a randomly chosen non-case

Model Performance



True Positive Rate (TPR) is a synonym for recall: $TPR = TP / (TP + FN)$

False Positive Rate (FPR) is defined as: $FPR = FP / (FP + TN)$

AUC	Discrimination
0.50	No better than chance
0.70–0.80	Acceptable
0.80–0.90	Excellent
> 0.90	Outstanding

AUC stands for "Area under the ROC Curve." and measures the entire two-dimensional area under the entire ROC curve.

Fitting in R: Base R (glm)

```
fit <- glm(outcome ~ x1 + x2, data = df, family = binomial(link = "logit"))
summary(fit)

# Odds ratios and CIs
exp(coef(fit))
exp(confint(fit))

# Predicted probabilities
predict(fit, type = "response")
```

`family = binomial()` is what switches `glm()` from OLS-style fitting to logistic regression
`summary()` gives coefficients, SEs, z-values, and p-values — plus null/residual deviance and AIC

Fitting in R: Tidymodels

```
library(tidymodels)

log_spec <- logistic_reg() |>
  set_engine("glm") |>
  set_mode("classification")

log_fit <- log_spec |>
  fit(outcome ~ x1 + x2, data = df)

tidy(log_fit, exponentiate = TRUE, conf.int = TRUE) # odds ratios + CIs
predict(log_fit, new_data = df, type = "prob")
```

Tidymodels standardizes the interface so it's swappable with other classifiers `tidy()` with `exponentiate = TRUE` gives odds ratios directly, no manual `exp()`

Common Pitfalls

- Complete/quasi-separation — a predictor perfectly (or near-perfectly) predicts the outcome → coefficients blow up, `glm()` warns "fitted probabilities numerically 0 or 1"
- Multicollinearity — inflates standard errors, same as in linear regression — check VIF
- Class imbalance — a rare outcome can yield a model that looks accurate but has poor sensitivity — always check the confusion matrix, not just accuracy
- Extrapolation — predictions near $p = 0$ or $p = 1$ are especially sensitive to model misspecification

Recap & What's next

- Logistic regression models log-odds as a linear function of predictors
 - Coefficients → odds ratios via exponentiation
 - Fit via maximum likelihood, not least squares
 - R workflow: `glm(family = binomial())` or tidymodels' `logistic_reg()`
- **Next session:** Hands-on - Predicting a binary outcome